

MBAV 443M - Statistical Modelling and AI-Driven Analytics

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Repository link: <https://github.com/justinthomasjoseph-hub/mbav443-final-exam>

1. Data Preprocessing

- **Missing values:** The dataset contains no conventional nulls, but several categorical fields (job, marital, education, default, housing, loan) use "unknown" as a placeholder for missing data. These were treated as missing and imputed with the column mode.
- **Leakage control:** The duration column (length of the last call) was dropped, since it is only known after a call concludes and would leak the outcome into the model, making it unusable for a genuine pre-call prediction.
- **Encoding:** The target y was label-encoded (no = 0, yes = 1). Remaining categorical predictors were one-hot encoded.
- **Class imbalance:** The target is imbalanced (~89% "no", ~11% "yes"). Both models were trained with `class_weight="balanced"` so they do not simply default to predicting the majority class.
- **Split:** Data was split 80/20 into training and test sets using stratified sampling to preserve the class ratio in both sets. Numeric features were standardized for Logistic Regression only (Decision Trees are scale-invariant).

2. Models Built

Two classification models were trained:

- Logistic Regression, a linear and highly interpretable baseline classifier.
- Decision Tree (max depth = 6), which is a non-linear model capable of capturing interactions between features, depth-limited to control overfitting.

3. Model Evaluation

Both models were evaluated on the held-out test set (20% of the data, 8,238 records) using Accuracy, Precision, Recall, F1-score, and ROC-AUC. Confusion matrices and ROC curves were also plotted.

Both models also perform almost identically on accuracy, but Logistic Regression edges ahead on ROC-AUC, while the Decision Tree has a marginally higher recall. Precision is low for both (~ 0.37), which is expected given the class imbalance and the choice to prioritize recall through balanced class weights.

5. Recommendation

I would recommend Logistic Regression for deployment, with the Decision Tree retained as a complementary, easily explainable tool for stakeholder communication.

- Discriminative power: Logistic Regression achieves the higher ROC-AUC (0.801 vs 0.792), meaning it ranks customers by subscription likelihood more reliably across all thresholds, the most robust metric here given the class imbalance.
- Business priority Recall: A missed genuine subscriber (false negative) represents a lost sale, while contacting a customer who declines only costs a phone call. Both models achieve comparably strong recall (~ 0.64 – 0.65), so this does not favour either model decisively, but Logistic Regression's better-calibrated probabilities make it more suitable for ranking leads.
- Interpretability and governance: Logistic Regression coefficients give a transparent, stable view of which factors raise or lower subscription odds that are useful for marketing strategy and regulatory/compliance review in a banking context.
- Operational cost: Logistic Regression is computationally inexpensive to retrain as new campaign data arrives and integrates easily into a live scoring pipeline for the call centre.
- Caveat: Precision remains low (~ 0.37) for both models, meaning a majority of flagged leads will not convert. This trade-off is acceptable if the bank prioritizes not missing subscribers, but the cost of extra calls versus the value of a new term deposit should be reviewed by the marketing team when setting the final classification threshold.